Technical Memo

TO: Technical Manager
FROM: Yarden Sasson
RE: Movie Review Sentiment Classification Model — Deployment Recommendation
DATE: March 22, 2026

Summary

I developed a binary text classifier to distinguish the sentiment associated with movie reviews using the Kaggle IMDB Movie Review dataset (50,000 reviews with sentiment). After testing Naive Bayes, Logistic Regression, SVM, and Random Forest with both TF-IDF and Count Vectorization, I recommend **Logistic Regression + TF-IDF** for production deployment.

Why This Model

When it comes to the sentiment analysis and classification, it is important to prioritize the F1 score. This will allow all the positive reviews to be flagged while making sure that the negative reviews will be flagged and not have too many missed detections while looking at a dataset that is already balanced. The recommended model achieved an F1 score of 0.89 which outperformed SVM with TF-IDF (F1 score of 0.88) and the Logistic Regression with CountVectorizer (F1 score of 0.87). The rest of the models are comparable in performance as they are within 0.03 of the best performing model, but this model (Logistic Regression + TF-IDF) still performed the best when looking at the F1 score. The testing of this model also showed that it can be easily generalized and applied to different datasets (including those outside of the domain the initial model was trained on) while maintaining higher performance statistics.

Key Limitation

The training data only has two columns (review, and sentiment) and does not have other factors that the company could use to make actionable decisions. Other pieces of information that could contribute to the insights generated from this model include movie/tv show title, genre, streaming method, and many more. I recommend either pull more information when scraping these reviews from IMDB in the future or find other lists and join them to get more information from the model that could lead to insights and action items for either current projects or upcoming projects. Without this, each individual review we are looking at will need to be read in order to understand the complete context and what needs to be improved.